

Group 13 of the periodic table houses the boron group.



Pellets of pure aluminum refined in a laboratory

The Boron Group

Except for the metalloid boron, all the elements in this group are metals. Aluminum is abundant in Earth's crust but takes a lot of energy to extract. Pure gallium is shiny and melts at just above room temperature. Indium forms compounds that are essential in many modern electronics, while thallium is a deadly poison. Scientists know very little about nihonium—the only artificial element in this group.

Aluminum

13

Al

This element is the most abundant metal in Earth's crust, in the form of aluminum oxides, and the third most abundant among all elements, after oxygen and silicon. A light and strong metal, aluminum has a huge number of applications in industry and construction, particularly in building aircraft.



EYEWITNESS

IRÈNE JOLIOT-CURIE

By bombarding a thin piece of aluminum with the nuclei of helium atoms, Irène Joliot-Curie (the daughter of Marie Curie) was the first to show it was possible to produce an element artificially—in this case, a radioactive form (isotope) of phosphorus.



Greenish-blue surface is soft and scratches easily.

Variscite

Many aluminum ores occur naturally. Variscite is formed when water rich in phosphates—naturally occurring compounds of phosphorus—reacts with aluminum-rich rocks. A striking green-blue colored mineral, variscite is sometimes mistaken for another gemstone, turquoise.

